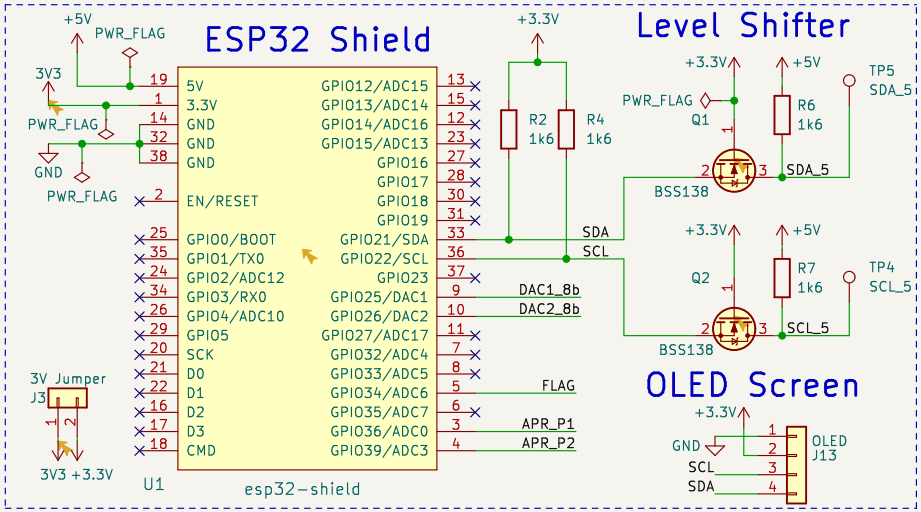
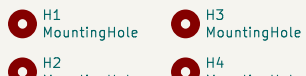
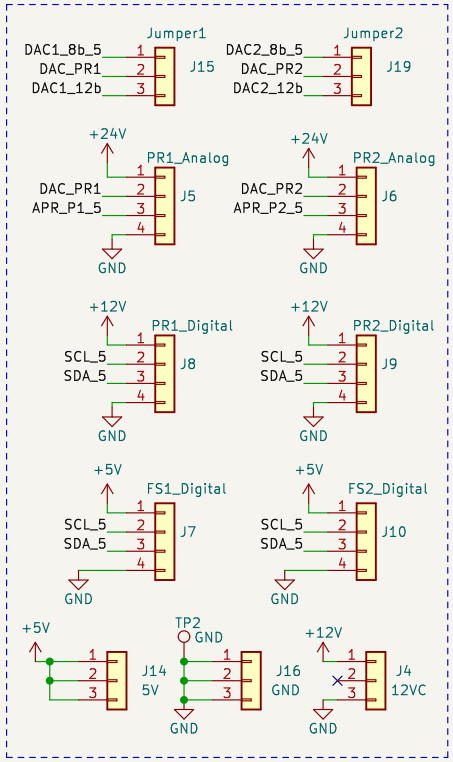
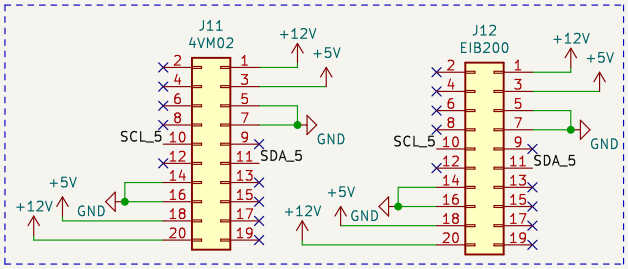




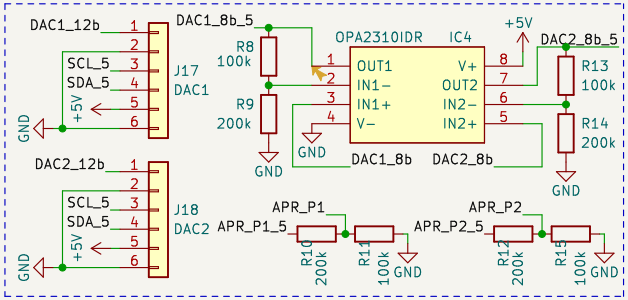
Pressure Regulator and Flow Sensor Interfaces



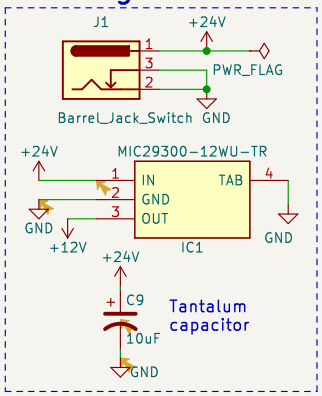
uProcess EIB/4VM Interfaces



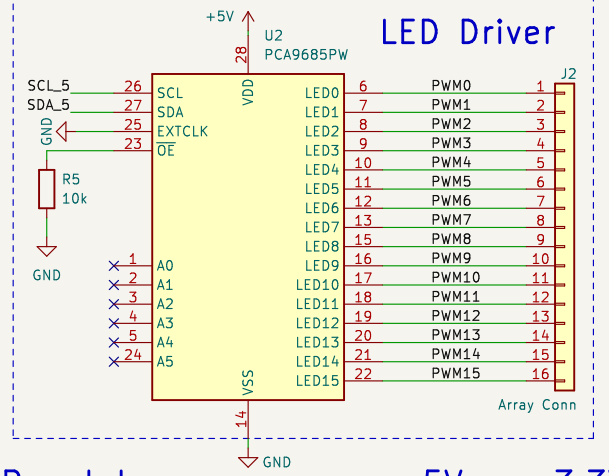
Analog Pressure Controller / 12-bit DACs



24V -> 12V Regulator



LED Driver



Functional Description of the PCB

INPUT Power: 24V (DC Barrel Jack)

MIC29300 Fixed Regulator – 24V -> 12V
MIC39101 Fixed Regulator – 12V -> 5V
TLV76733 Fixed Regulator – 5V -> 3.3V
OPA2310 Operational Amps – (0-3.3)V -> (0-5)V

I2C (5V/3.3V) Connections:

Labsmith EIB200 and 4VM02 Modules (5V)
Sparkfun MCP4725 12-bit DAC Modules (5V)
Sensirion SLI-0430FMK Flowrate Sensor (5V)
CG49 Digital Pressure Controllers (5V)
OLED Display (3.3V)
BSS138 Level Shifter Circuit – 3.3V -> 5V

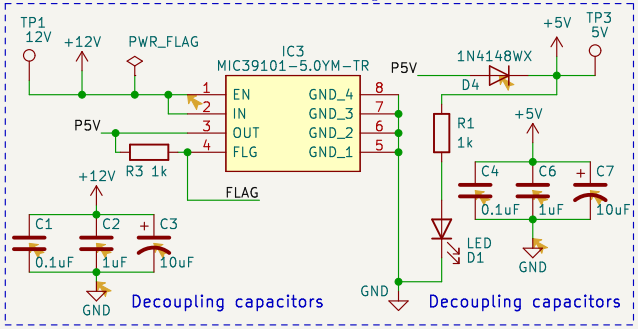
Analog (5V) Connections:

Parker (990-005001-015) Pressure Regulator (w/ Internal Vent)

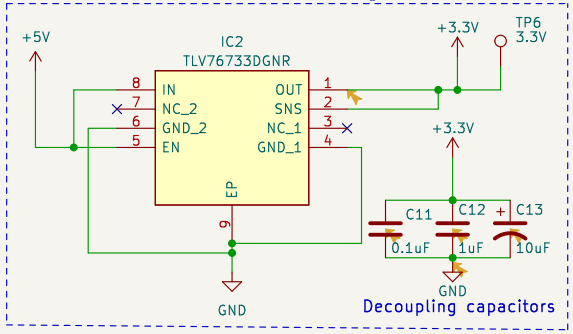
Microcontroller Module:

ESP32-WROOM-32E (3.3V) Dev Kit with 2.4 GHz WiFi + Bluetooth® + Bluetooth LE module

12V -> 5V Regulator



5V -> 3.3V Regulator



UBC SoC Lab Biosensors Research Group	
Sheet: /	
File: Fluidics_PCB_v4.kicad_sch	
Title: Microfluidics Control System PCB v4	
Size: A4	Date: 2023-05-24
KiCad E.D.A. kicad (6.0.9-0)	**Rev: Rev4**
	Id: 1/1